NB-234: Liquid phase calibration of O2 robust probe in H2/O2 photoreactor

Date: 2025-06-11
Tags: O2 Calibration Future NB Firesting Degassing O2 sensor H2/O2 reactor
Category: Prep work
Status: Done
Created by: Nadzeya Brezhneva

Reaction scheme/sample structure

/

ChemDraw File (linked): /

Literature/reference experiments

Literature	/
Reproduction	/
Similar experiments	Prep work - NB-233: Liquid phase calibration of O2 robust oxygen sensor (Firesting) SrTiO3 - NB-AE-229: Liquid phase H2 and O2 measurements with Unisense H2 sensor, Firesting probe in irradiated Al:SrTiO3 RhCrOx (NB-162-4, 0.8 mg/mL), 365 nm, 50 mW, 1 h, degassing

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	$Mass_{theo}$ [mg]	$Mass_{exp}$ [mg]	Molar mass [g/mol]	Density (g/ml)	Volume
milli-Q H2O	7732-18-5	/	/	/	/	/	18	0.998	25 mL
balloon with compressed air	/	/	/	/	/	/	/	/	ca. 2 L (very rough estimation)

Irradiation Parameters

	Name
Used Set-up (select one)	Equipment - Manual irradiation setup Irradiation setup #2

Procedure/observations

Date	Time	Step	Observations	Pictures

11.06.2025	ca. 16:00-20	Assembling the photoreactor for H2/O2 simultaneous detection. Robust O_2 probe - NS14/GL14 connector, 3 mm BOLA fitting PT 100 - NS14/GL14 connector, 4 mm BOLA fitting Before assembling, BOLA compartments were cleaned with compressed air. Initially, BOLA compartments were assembled on the appropriate sensor (see the configuration of BOLA fitting above), afterwards NS14/GL14 adapter was set and tightly screwed. Next, the grease was put on the NS14 part of the adapter and placed in the NS14 neck of the photreactor. The BOLA/adapter/sensor assembly was carefully rotated for 30° several times to distribute grease evenly, afterwards the configuration was fixed with plastic clamp.	O2 sensor - in the central neck of the reactor, PT100 - one of the side neck, stopper and septum on two other necks	20250611_162518-before purging with air.jpg
	ca. 16:22	Filling the reactor with 30 mm*2mm PTFE stirrer and filling with 25 mL of water via graduated cylinder.	/	/
	ca. 16:25	Filling balloon with compressed air (CA).	/	/
	16:30	Start bubbling the system with air via septum.	/	20250611_164257-during air bubbling.jpg
	16:44-45	Taking upper point calibration	256 µmol/L	20250611_164513-during calibration.jpg
	16:46	Removing the balloon with CA.	/	/
	16:48-50	Preparing the setup for degassing with argon, immersing PTFE cannula into the reactor via the valve.	/	/
	16:51	Start degassing.	/	/
	17:12	Taking 0% calibration point during degassing	-0.07 mbar	
	17:13	Calibartion was applied and saved in a new file.	/	20250611-BOLA-fitting-liquid-phase-trace-oxygen-sensor.ini
	17:13-15	Deassembling the setup for degassing.	/	/
	17:15	Start logging to check sealing of the reactor (rotation speed 500 rpm).	17:22 increasing rotation speed up to 600 rpm (O2 sensor became in the funnel in the liquid), decreasing to the value 540 rpm. O2 level in the range 0.1-0.3 µmol/L.	Not saved.
		Important note: Current configuration of the setup (*O2 sensor in the center of the reactor, PT100 on the side, stopper and septum on the others neck, 25 mL of liquid inside the reactor, the sensors are in the liquid phase and immersed ca. 4-5 mm deep into the liquid*) was assembled with the following parameters (position of BOLA fitting on the sensor): BOLA-robust O2 probe ---> 63 mm from the blue tape on the top of the sensor BOLA-PT100 ---> 87 mm from the white cable/sensor connection		

Results

Liquid pase calibration of O2 robust probe (Pyroscience) in photoreactor for H2/O2 simultaneous detection was performed in combination with PT100 sensor. New calibration file was created.

Future recommendations

Old procedure	Problem	Suggested new procedure
/	/	Use the following procedure for recording data during calibration procedure: Start logging --> Bubbling with air or degassing --> Stop logging ---> Start calibration ---> Take calibration point Record logging after calibration to check sealing of the system.

Linked experiments

Prep work - NB-233: Liquid phase calibration of O_2 robust oxygen sensor (Firesting)

$SrTiO_3$ - NB-AE-229: Liquid phase H_2 and O_2 measurements with Unisense H_2 sensor, Firesting probe in irradiated Al:$SrTiO_3$ $RhCrO_x$ (NB-162-4, 0.8 mg/mL), 365 nm, 50 mW, 1 h, degassing

$SrTiO_3$ - NB-231: Gas phase H_2 and O_2 measurements with Unisense H_2 sensor, Firesting O_2 cap in irradiated Al:$SrTiO_3$ $RhCrO_x$ (NB-162-4, 1 mg/mL), 365 nm, 50 mW, 1 h, degassing

Attached files

20250611_164513-during-calibration.jpg
sha256: c023005ff348af0aa3dda0d8d613a4db5455b787832c5ff2ad9a7a8f52af2030

20250611_162518-before-purging-with-air.jpg
sha256: ec4f8f906b7c82a33672181f1858eec71978dd3a950fbb813f9b3185db66fc73

20250611_170703-during-degassing-with-argon.jpg
sha256: 4745a33428988ef6c2f4b40ee8723fed24e34875b00d0fa15c9c123afa54b8e7

20250611_164257-during-air-bubbling.jpg
sha256: c06bde6eabb7bc86a0a4e5fd203cf8bdc5a4094a1c71dc194b5f16ee40dbcadc

20250611-BOLA-fitting-liquid-phase-trace-oxygen-sensor.ini
sha256: 8163400e6cadf8db505063df4c390472c98d313d2b52597b7b1c4200a3821831

Unique eLabID: 20250611-94f6fdbf0bfc9b490c3215cb35cfef5d5cc07676
Link: https://elab.water-splitting.org/experiments.php?mode=view&id=2250